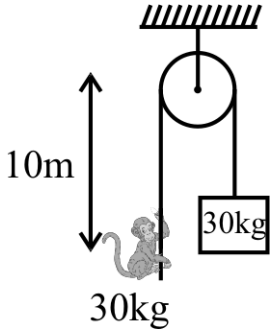
PART-1 : PHYSICS

SECTION-I

1) In the figure shown the monkey is balanced by counter weight of same mass. The monkey starts to climb the rope with an acceleration of 2 m/s^2 w.r.t. rope. Neglecting size of monkey find the time after which he reaches the pulley :



(A) $\sqrt{10}$ sec

(B) $2\sqrt{5}$ sec

(C) $2\sqrt{10}$ sec

(D) He can't reach pulley

2) A semicircular loop of radius r is placed on the axis of a current carrying circular ring of radius R (R >> r), at a distance $R\sqrt{3}$ from the centre of the ring. The plane of the loop is making an angle of 53° with the plane of the ring. Find the coefficient of mutual induction between them :-

(A) $\frac{\mu_0 \pi r^2}{40R}$

(B) $\frac{3\mu_0 \pi r^2}{160R}$

(C) $\frac{3\mu_0 \pi R^2}{160r}$

(D) None of these

3) **Statement–1** : For ampere's law $\oint \vec{B}.\vec{d\ell}$ = $\mu^0 I$, B also includes the contribution from current not enclosed by the loop.
and
Statement–2 : A moving charge produces magnetic field, but no electric field.

(A) Statement–1 is True, Statement–2 is False.

(B) Statement–1 is False, Statement–2 is True.

(C) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1

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(D) Statement–1 is True, Statement–2 is True ; Statement–2 is NOT a correct explanation for Statement–1

4) A rubber band (two parallel strands of elastic material forming a closed loop) has an effective spring constant of 20 N/m. If the band is cut in one place such that it now forms a single long strand of elastic material, what is its new effective spring constant (in N/m)?

(A) 5

(B) 6

(C) 6.5

(D) 7

5) The magnetic susceptibility of a paramagnetic substance at – 73°C is 0.0060, then its value at – 173° C will be :

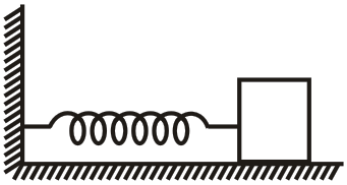
(A) 0.0030

(B) 0.0120

(C) 0.0180

(D) 0.0045

6) In the situation shown, the block can execute free oscillation (no damping) with angular frequency ω_1. In presence of weak damping, it executes damped SHM with angular frequency ω_2. When it is subjected to a sinusoidal force, it executes forced oscillation with maximum amplitude at angular frequency ω_3 (assume damping is present), then :



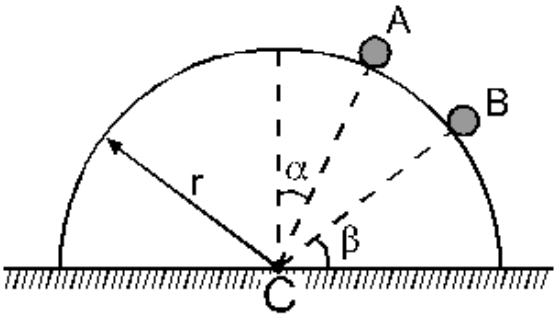
(A) $\omega_1 > \omega_2 > \omega_3$

(B) $\omega_1 > \omega_2 = \omega_3$

(C) $\omega_1 = \omega_2 = \omega_3$

(D) $\omega_1 > \omega_3 > \omega_2$

7) A particle initially at rest starts moving from point A on the surface of a fixed smooth hemisphere of radius r as shown. The particle looses its contact with hemisphere at point B. C is centre of the hemisphere. The equation relating α and β is



(A) $3 \sin \alpha = 2 \cos \beta$

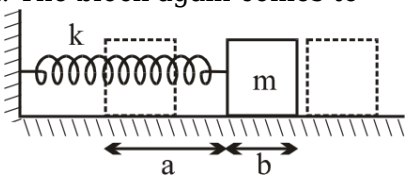
(B) $2 \sin \alpha = 3 \cos \beta$

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(C) $3 \sin \beta = 2 \cos \alpha$

(D) $2 \sin \beta = 3 \cos \alpha$

8) The spring is compressed by distance 'a' and the system is released. The block again comes to rest when the spring is elongated by a distance b. During this process



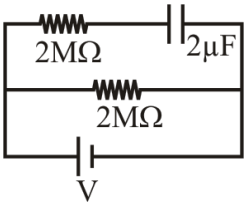
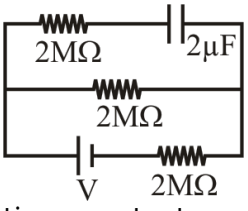
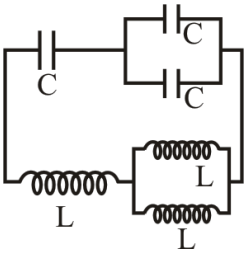
(A) work done by the spring on the block = $\frac{1}{2}k(a-b)^2$

(B) work done by the spring on the block = $\frac{1}{2}k(a^2+b^2)$

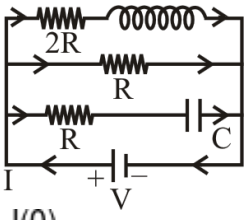
(C) coefficient of friction = $\frac{k(a-b)}{2mg}$

(D) coefficient of friction = $\frac{k(a+b)}{2mg}$

9) Match the physical quantity of List-I with value given in List-II.

	List-I		List-II
(P)	 time constant τ = ___ sec	(1)	100
(Q)	 time constant τ = ___ sec	(2)	1.33
(R)	 C = 100 μF, L = 1H angular frequency ω = ... rad/sec	(3)	4

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(S)	 $\frac{I(0)}{I(\infty)}$ = ... I(0) and I(∞) are current drawn from the battery at t = 0 and t → ∞	(4)	6

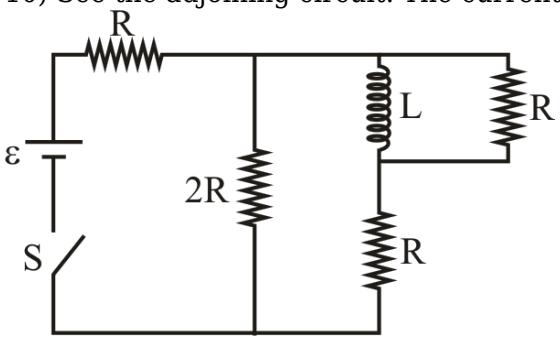
(A) P→2, Q→1, R→4, S→3

(B) P→3, Q→1, R→4, S→2

(C) P→3, Q→4, R→1, S→2

(D) P→4, Q→2, R→3, S→1

10) See the adjoining circuit. The current through the cell just after closing the switch S is :

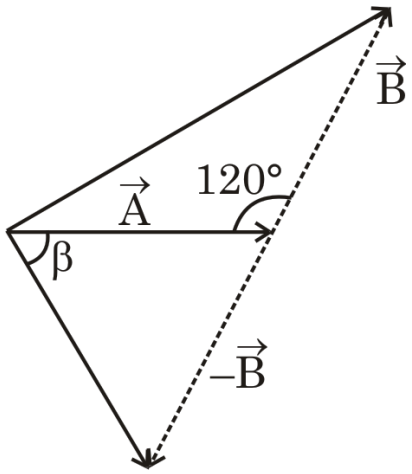


(A) $\frac{\varepsilon}{R}$

(B) $\frac{2\varepsilon}{R}$

(C) $\frac{\varepsilon}{3R}$

(D) $\frac{\varepsilon}{2R}$



11) The angle between vector $(\vec{A})$ and $(\vec{A}-\vec{B})$ is :-

(A) $\tan^{-1}\left(\frac{-\frac{B}{2}}{A-B\frac{\sqrt{3}}{2}}\right)$

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(B) $\tan^{-1}\left(\frac{A}{0.7B}\right)$

(C) $\tan^{-1}\left(\frac{\sqrt{3}B}{2A-B}\right)$

(D) $\tan^{-1}\left(\frac{B\cos\theta}{A-B\sin\theta}\right)$

12)

The wavelength associated with a moving particle depends upon p^{th} power of its mass m, q^{th} power of its velocity v and r^{th} power of Planck's constant h. Then, the correct set of values of p, q and r is (E = hν, E is the energy of a wave, ν is frequency & h is planks constant)

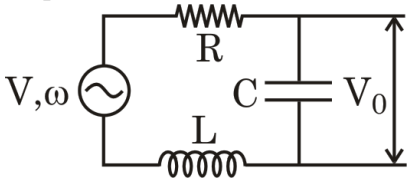
(A) p = 1, q = –1, r = 1

(B) p = 1, q = 1, r = 1

(C) p = –1, q = –1, r = –1

(D) p = –1, q = –1, r = 1

13) In the circuit shown in figure, V_0 is the voltage across capacitor. V is the ac source amplitude and ω is angular frequency of source. If ω is very large :-



(A) $\frac{V_0}{V}\propto\omega$

(B) $\frac{V_0}{V}\propto\omega^2$

(C) $\frac{V_0}{V}\propto\omega^{-1}$

(D) $\frac{V_0}{V}\propto\omega^{-2}$

14) A long solenoid is made of a superconducting wire (have zero resistance) carrying a current I_0. The solenoid is slowly stretched so that its cross section does not change but its length changes from L_0 to L_1. What is the new current in the solenoid assuming the solenoid is still tightly wound?

(A) $\frac{L_0 i_0}{L_1}$

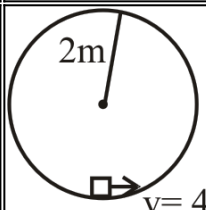
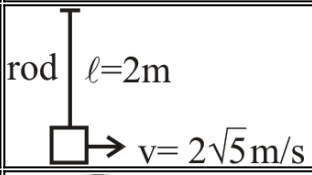
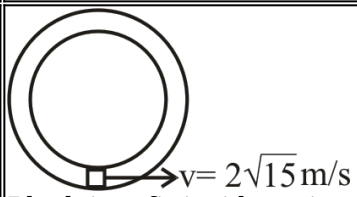
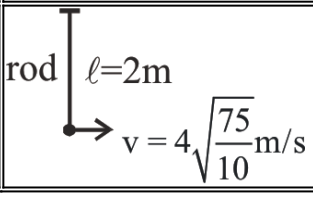
(B) $\frac{L_1 i_0}{L_0}$

(C) $\frac{L_0^2 i_0}{L_1^2}$

(D) $\frac{L_1^2 i_0}{L_0^2}$

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15) List-I gives some situations where a block is given velocity to perform vertical circular motion and list-II give information about motion of block and force acting on the blocks. Match the correct order.

	List-I		List-II
(P)	 Block is given velocity v in a smooth circular loop of radius 2m	(1)	Block completes circular motion
(Q)		(2)	Normal reaction at highest point is zero
(R)	 Block just fit inside a circular tube of radius 2m given velocity as shown	(3)	Tension at highest point is zero.
(S)		(4)	Normal reaction at highest point is radially outwards.

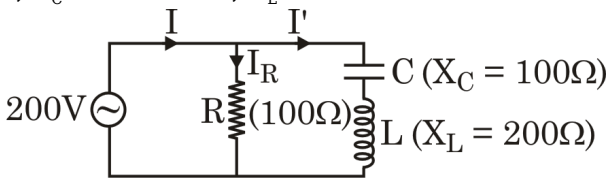
(A) P→3, Q→2, R→1, S→4

(B) P→2, Q→3, R→4, S→1

(C) P→2, Q→4, R→3, S→1

(D) P→1, Q→3, R→4, S→2

16) In the circuit diagram shown, X_C = 100 ohms, X_L = 200 ohms & R = 100 ohms. The effective current through the source is :-



(A) 2 A

(B) $2\sqrt{2}$ A

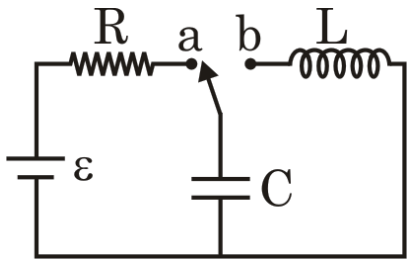
(C) 0.5 A

(D) None of these

17) The switch in the circuit pictured is in position a for a long time. At t = 0 the switch is moved from a to b. The current through the inductor will reach its first maximum after moving the switch in

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a time:



(A) $2\pi\sqrt{LC}$

(B) $\frac{1}{4}\sqrt{LC}$

(C) $\frac{\pi}{2}\sqrt{LC}$

(D) $\pi\sqrt{LC}$

18) A man is running up the inclined plane (making an angle α from the horizontal) with a constant velocity. Rain is falling perpendicular to the incline plane. During motion man observes that rain is falling in vertical direction, when man doubles his velocity he observes that rain is making an angle α from the incline. Value of α is :-

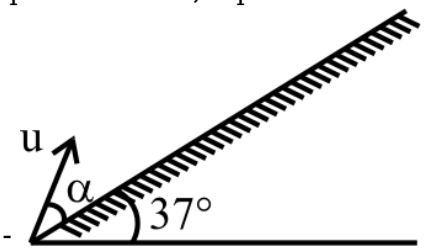
(A) $\tan^{-1}(0)$

(B) $\tan^{-1}(\sqrt{2})$

(C) $\tan^{-1}(1/\sqrt{2})$

(D) $\tan^{-1}(1/2)$

19) As shown particle is projected up the inclined, if particle strikes perpendicularly to the inclined plane then find value of α :-



(A) $\tan^{-1}\left(\frac{4}{3}\right)$

(B) $\tan^{-1}\left(\frac{3}{4}\right)$

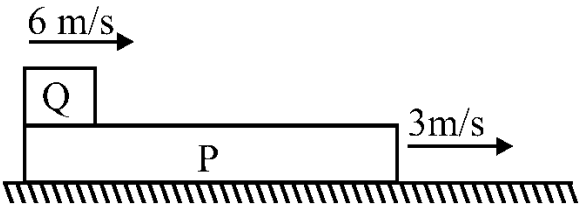
(C) $\tan^{-1}\left(\frac{3}{2}\right)$

(D) $\tan^{-1}\left(\frac{2}{3}\right)$

20) A plank P and block Q are arranged as shown on a smooth table top. They are given velocities 3 m/s and 6 m/s respectively. The length of plank is 1m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. The velocity of plank then is (coefficient of friction between plank and block is 0.3 and mass of plank is double of block

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and no friction between plank & ground) :-



(A) 4 m/s

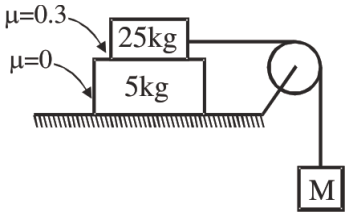
(B) 5 m/s

(C) 4.5 m/s

(D) zero

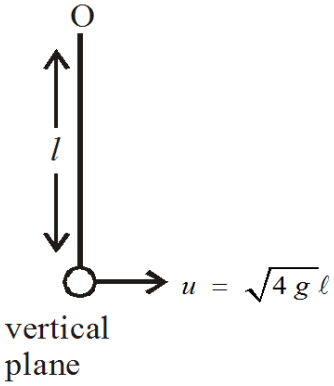
SECTION-II

1) Two blocks of masses 25 kg and 5 kg are placed on a horizontal table as shown in figure. A massless string passes over a frictionless and massless pulley whose one end is connected to 25 kg block and other end is connected to block M. The coefficient of friction between two blocks is μ = 0.3 and between the 5 kg block and ground is zero. The system is released from rest. If M = 45 kg and the force of friction between two blocks is f, then find the value of f/6 (in N). (Take g = 10 m/s^2)



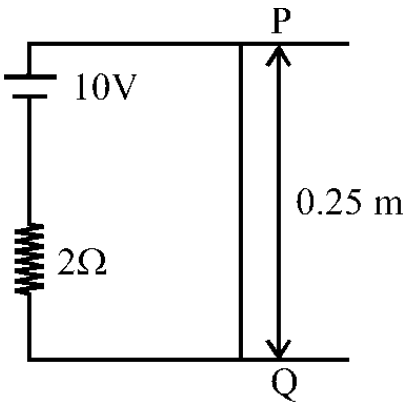
2) A solid cylinder of radius 1m and mass 10kg is free to rotate about its axis. A torque of 5πNm is applied to it for a period of 3sec and then it is suddenly reversed in direction without changing the magnitude. The angle rotated by the cylinder in time interval 6 to 10 sec is θ radian. If θ = nπ, fill n.

3) A pendulum bob hinged at O is given velocity $u = \sqrt{4g\ell}$ from bottommost point as shown. If ℓ = 54cm, find maximum height (in m) reached by bob from point of projection.



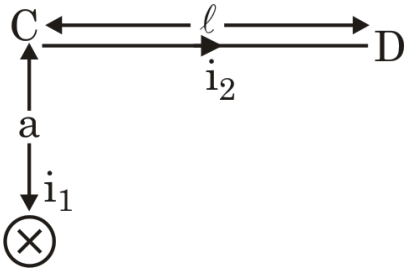
4) A metal wire PQ slides on parallel metallic rails having separation 0.25 m, each having negligible resistance. There is a 2Ω resistor and 10V battery as shown in figure. There is a uniform magnetic field directed into the plane of the paper of magnitude 0.5 T. A force of 0.5 N to the left is required to keep the wire PQ moving with constant speed to the right. With what speed (in m/s) is the wire PQ

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moving ? (Neglect self inductance of the loop)

5) An infinitely long straight wire carries a current i_1, as shown in figure. The total force on another wire CD of length ℓ which is placed so that c is at a distance a from the current carrying wire is



$\frac{\mu_0 i_1 i_2}{N\pi} \ln\left(\frac{a^2+\ell^2}{a^2}\right)$. Then find value of N.

PART-2 : CHEMISTRY

SECTION-I

1) Select the **CORRECT** option from the given below.

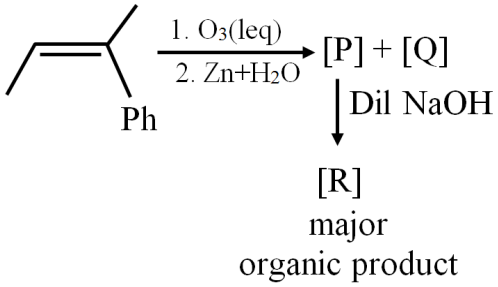
(A) Sodium salt of propionic acid on decarboxylation with sodalime gives propane as a major product

(B) Benzoic acid is more acidic than acetic acid

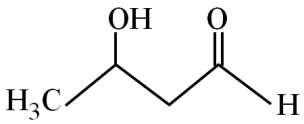
(C) Methanoic acid $\xrightarrow[2.\ H_2O]{1.\ Cl_2/\text{Red phosphorus}}$ CH_2COOH | Cl

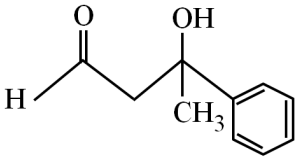
(D) All

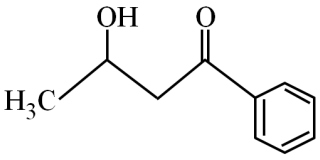
2) Identify the major organic product [R] in the given reaction sequence.

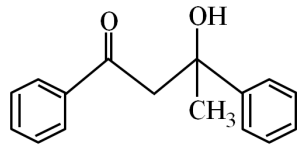


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(A) 

(B) 

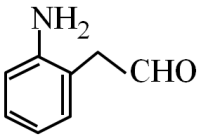
(C) 

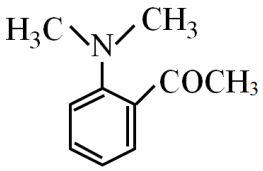
(D) 

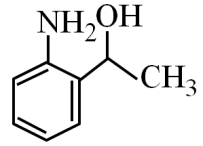
3) The test performed on compound X and their inferences are :

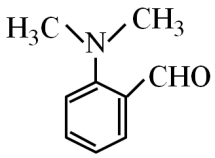
	Test	Interference
(a)	2,4-DNP test	No coloured precipitate
(b)	Iodoform test	Yellow precipitate
(c)	Azo-dye test	Dye formation

Compound 'X' is

(A) 

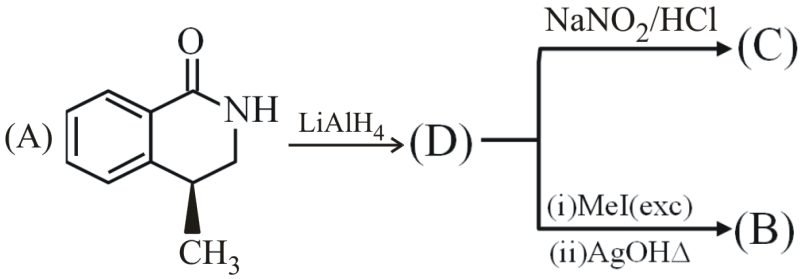
(B) 

(C) 

(D) 

4) With reference to the scheme given below, which of the following statement (s) is/are true about compounds (A), (B), (C) and (D) ?

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(a) Compound (A) releases ammonia when boiled with concentrated NaOH solution.
(b) Compound (C) is an yellow oily liquid and is chiral.
(c) Compound (D) exhibit geometrical isomerism.
(d) Compound (D) on reaction with $PhSO_2Cl$ give a product soluble in aq. NaOH

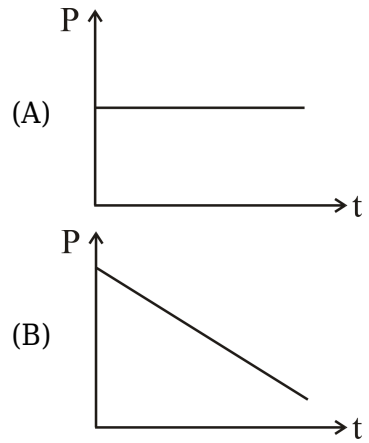
(A) b, c
(B) a, b
(C) a, b, c, d
(D) a, b, d

5) For any reaction : A(g) → P(s) following observation is made :

Time (min)	Partial pressure of A (in bar)
0	20
20	10
60	5

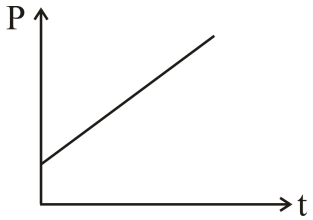
Order of the reaction may be -
(A) –1
(B) 0
(C) 1
(D) 2

6) If decomposition reaction : A(g) → B(g) + C(g) follows zero order kinetics, then the graph of total pressure (P) against time t will be

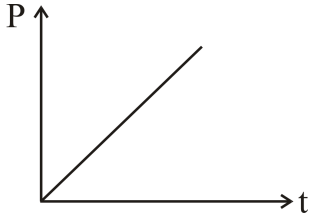


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(C)



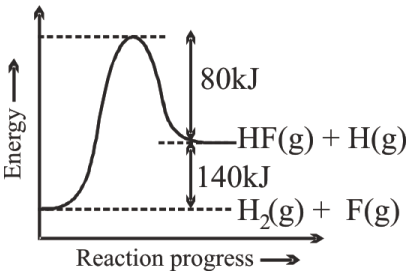
(D)



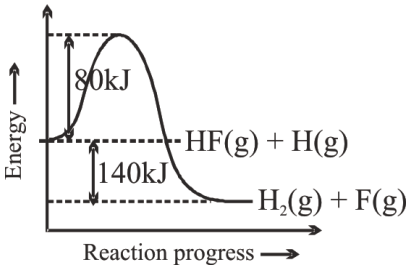
7)

For an elementary reaction : $H_2(g) + F(g) \rightarrow HF(g) + H(g)$; $\Delta H = -140$ kJ/mol
Activation energy of the forward reaction 80 kJ/mol. The correct energy diagram is :

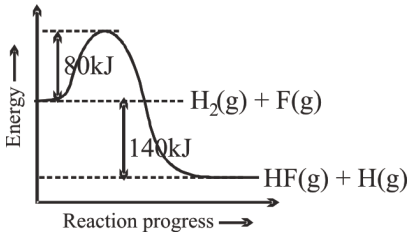
(A)



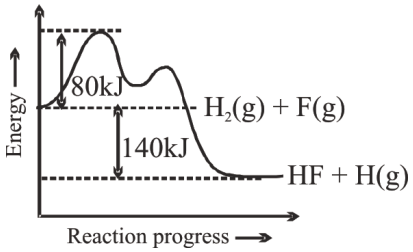
(B)



(C)



(D)



8) For most reactions, rate of a reaction increases about two times when its temperature is raised by 10 K. This fact is due to

(A) increase in number of collisions per unit volume

(B) increase in number of molecules having energy E_A or more

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(C) lowering of activation energy

(D) lowering of pre-exponential factor in Arrhenius equation

9) The reaction mechanism for the reaction : P → R is as follows :

$$P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2Q(\text{test})$$

$$2Q + P \xrightarrow{k_3} R(\text{slow})$$

The rate law for the reaction (P → R) is :

(A) $k_1[P]\,[Q]$

(B) $k_1k_2[P]$

(C) $\frac{k_1k_3[P]^2}{k_2}$

(D) $k_1k_2[a]$

10) **Statement-I** : Product can form only when the required orientation and energy conditions of collision are met.

Because

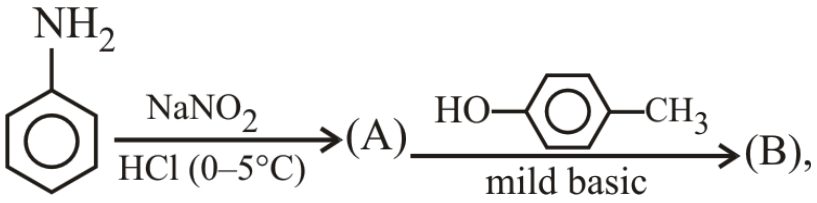
Statement-II : All collisions between reactants yield the desired product.

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

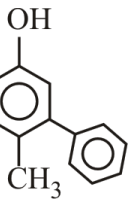
(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I

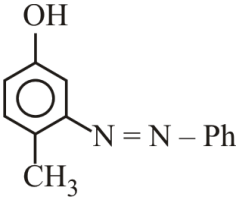
(C) Statement-I is true, Statement-II is false

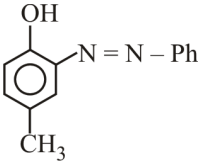
(D) Statement-I is false, Statement-II is true

11) 

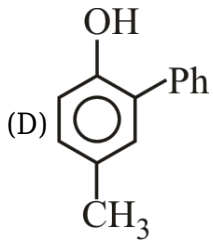
Product (B) is

(A) 

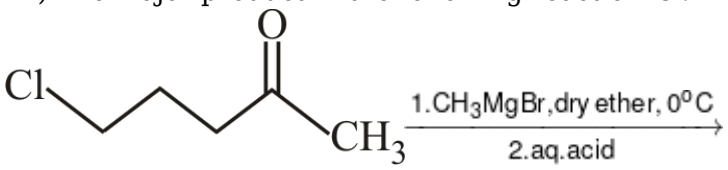
(B) 

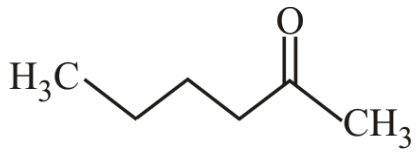
(C) 

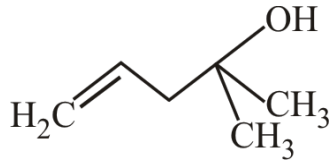
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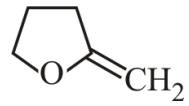
(D) 

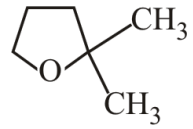
12) The major product in the following reaction is :



(A) 

(B) 

(C) 

(D) 

13) Among the following, the correct indicated property of the isoelectronic pair is

(A) O_2 and NO^- ; same bond order

(B) $Ni(CO)_4$ and $[Fe(CO)_4]^{2-}$; same metal carbon bond length

(C) CO_3^{2-} and NO_3^- ; same bond length

(D) $Fe(CO)_5$ and $Cr(CO)_6$; different magnetic behaviour

14) Which of the following option is **CORRECT** for $[FeF_6]^{3-}$ & $[CoF_6]^{3-}$?

(A) Both have same magnetic moment

(B) Both are high spin complex

(C) Both have same electrons in t_{2g} orbitals.

(D) None of these

15) Which of the following statement is **CORRECT** for $[NiCl_2(NH_3)_2]$ (spin magnetic moment = 0)

(A) It is optically active complex

(B) It is high spin complex

(C) It is tetrahedral complex

(D) It has two stereo isomer

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16) Which of the following order is **CORRECT** for $\Delta_0/\Delta t/\Delta_{sp}$?

(A) $[Fe(SCN)_6]^{3-} > [Fe(C_2O_4)_3]^{3-}$

(B) $[Co(H_2O)_3F_3] > [Co(H_2O)_6]^{3+}$

(C) $[NiF_6]^{-2} > [NiF_6]^{4-}$

(D) $[Fe(H_2O)_6]^{2+} > [Fe(H_2O)_6]^{3+}$

17) If the bidented ligand ethane 1-2diamine(en) is progressively added to $[Ni(H_2O)_6]^{+2}$ in the molar ratio
en :$[Ni(H_2O)_6]^{+2}$; 1 : 1, 2 : 1, 3 : 1 then complex form are (P) (Q) and (R) respectively.
Select incorrect statement for (P) (Q) and (R).

(A) All have same oxidation number and same coordination number of metal

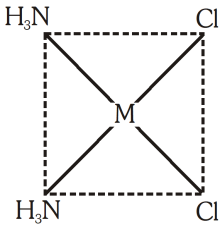
(B) All have Ni–N linkage(s) but NO linkage isomer(s)

(C) All have Ni–O linkage(s) and geometrical isomer(s).

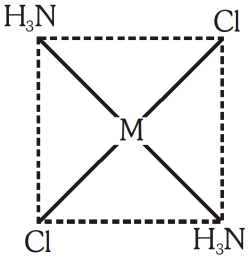
(D) All have EAN = 38 of metal and complex having chelate ring(s).

18) Which of the following isomers of $[M(NH_3)_2Cl_2]$ would react with silver oxalate ($Ag_2C_2O_4$):

(A)



(B)



(C) Both

(D) None

19)

List-I (Complex ions)		List-II (Hydridisation with magnetic moment)	
(P)	$[Cr(NH_3)_6]^{3+}$	(1)	dsp^2, $\mu = 0$
(Q)	$[FeF_6]^{3-}$	(2)	d^2sp^2, $\mu = 0$
(R)	$[Co(NH_3)_6]^{3+}$	(3)	d^2sp^3, $\mu = \sqrt{15}$
(S)	$[Pt(NH_3)_2(Cl)_2]$	(4)	sp^3d^2, $\mu = \sqrt{35}$

(A) P → 3 ; Q → 4 ; R → 2 ; S → 1

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(B) P → 2 ; Q → 4 ; R → 3 ; S → 1

(C) P → 1 ; Q → 3 ; R → 4 ; S → 2

(D) P → 4 ; Q → 3 ; R → 2 ; S → 1

20) **Assertion :** Phenoxide ion is more stable than carboxylate ion hence phenol is more acidic than carboxylic acids.
Reason : The negative charge is delocalized over two electronegative oxygen atoms in carboxylate ion.

(A) Both assertion and reason are true and the reason is a correct explanation of assertion.

(B) Both assertion and reason are true and the reason is a incorrect explanation of assertion.

(C) Assertion is true but the reason is false.

(D) Assertion is false but the reason is true.

SECTION-II

1) Rate of a reaction increases 512 times when temperature is increased from 10^0C to 100°C. What is the temperature coefficient of reaction ? Assume that on increasing the temperature by 10°C, the rate always increases by the same factor.

2) In the decomposition of ammonia, it was found that at initial 50 torr pressure, $t_{1/2}$ was 3.64 hour, while at initial 100 torr pressure, $t_{1/2}$ was 1.82 hours. Then, the order of reaction would be :

3) Among the given reactions in how many reactions number of carbon atoms is same in starting material and final product (note : consider major organic product in each case)

1) $CH_3CH_2I \xrightarrow{NaCN} \xrightarrow[\text{partial hydrolysis}]{OH/H_2O} \xrightarrow{NaOH+Br_2}$ [P] major product

2) $C_6H_5NO_2 \xrightarrow{Fe/HCl} \xrightarrow[273K]{NaNO_2+HCl} \xrightarrow[\Delta]{H^+/H_2O}$ [Q] major product

3) $C_2H_5Br \xrightarrow{KCN} \xrightarrow[H_2O]{LiAlH_4} \xrightarrow[0^oC]{HNO_2}$ [S] maor product

4) $CH_3CO_2H \xrightarrow[\Delta]{NH_2} \xrightarrow{NaOBr} \xrightarrow{NaNO_2+HCl}$ [T] major product

5) $C_6H_5NO_2 \xrightarrow{Fe/HCl} \xrightarrow[273K]{NaNO_2+HCl} \xrightarrow[\Delta]{CuCN/KCN} \xrightarrow[H_2O]{LiAlH_4}$ [U] major product

6) $C_6H_5NO_2 \xrightarrow{Fe/HCl} \xrightarrow[273K]{NaNO_2+HCl} \xrightarrow[H_2O]{H_3PO_2}$ [V] major product

4) In the following reaction, the number of monochloro derivatives possible for the compound 'D' is :

$$CH_3CN \xrightarrow{H_2O/H^+} A \xrightarrow{Ca} B \xrightarrow[\Delta]{\text{dry distilation}} C \xrightarrow{Zn-Hg+HCl} D$$

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5) Total number of ligand(s) which forms five membred ring & also contains chiral centre(s).
pn, en, tn, bn, DMG^{-1}, $acac^{-1}$, nta^{-1}

PART-3 : MATHEMATICS

SECTION-I

1) Number of integral values of λ for which $x^2 - 2\lambda x < 41 - 6\lambda \ \forall \ x \in (1, 6]$, is -

(A) 10
(B) 11
(C) 12
(D) 13

2) If $(y^2 - 5y)(x^2 + 2x + 4) < 2$, for all $x \in R$, then number of integers in the range of y is

(A) 1
(B) 2
(C) 3
(D) 6

3) **Assertion (A) :** The least value of $\left(\frac{\tan^2 A + \tan A + 1}{\tan A}\right)$ is 3, where $A \in \left(\pi, \frac{3\pi}{2}\right)$.
Reason (R) : A.M. $\geq$ G.M. for all positive numbers.

(A) **A** is correct but **R** is not correct
(B) **A** is not correct but **R** is correct
(C) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**
(D) Both **A** and **R** are correct and **R** is the correct explanation of **A**

4) Given all terms of an A.P. are positive integers, third term is 8 and sum of first 15 terms lies between 191 and 196, then sum of first 20 terms is -

(A) 310
(B) 330
(C) 290
(D) 340

5) Let $V_{n+1} = 3V_n - 2V_{n-1} \ \forall \ n \in N$ and $V_0 = 2$, $V_1 = 3$, then value of V_6 is

(A) 65
(B) 33
(C) 17
(D) 36

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6) Number of rational terms in the expansion of $\left(3^{\frac{1}{8}}+5^{\frac{1}{3}}\right)^{400}$ is

(A) 17

(B) 20

(C) 102

(D) 150

7) Let $(1 + x + x^2)^{20}(2x + 1) = a^0 + a_1x^1 + a_2x^2 + ... + a_{41}x^{41}$, then $\frac{a_0}{1} + \frac{a_1}{2} + ... + \frac{a_{41}}{42}$ is equal to -

(A) $\left(\frac{2^{21}-1}{21}\right)$

(B) $\left(\frac{3^{21}-1}{21}\right)$

(C) $\left(\frac{2^{20}-1}{20}\right)$

(D) $\left(\frac{3^{20}-1}{20}\right)$

8) Total number of words which can be formed by taking all the letters of the word 'ROORKEE' such that among the alike letters only 'RR' is always together

(A) 60

(B) 84

(C) 96

(D) none of these

9) Consider the quadratic trinomial $P(x) = 2x^2 - 10px + 7p - 1$, where p is a parameter, find the range of 'p' in the following conditions given in.
Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(A)	If both roots of P(x) = 0 are confined in (–1,1), then	(I)	$\left(\frac{1}{3},\infty\right)$
(B)	Exactly one root of P(x) = 0, lies in (–1,1).	(II)	ϕ
(C)	Both roots of P(x) = 0 are greater than 1	(III)	$\left(-\frac{1}{17},\frac{1}{3}\right)$
(D)	One root of P(x) = 0 is greater than 1 and other roots of P(x) = 0 is less than 1.	(IV)	$\left(-\infty,-\frac{1}{17}\right]\cup\left[\frac{1}{3},\infty\right)$

(A) (A) – III, (B) – II, (C) – I, (D) – IV

(B) (A) – III, (B) – IV, (C) – II, (D) – I

(C) (A) – III, (B) – II, (C) – IV, (D) – I

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(D) (A) – I, (B) – II, (C) – III, (D) – IV

10) 3 boys B_i, i = 1, 2, 3 and 6 girls G_i, i = 1, 2, ..., 6 are to be seated in a row. Number of ways they can be seated so that B_1, B_2 are separated and G_1, G_2 are also separated equal to

(A) $5 \lfloor 8$
(B) $44 \lfloor 7$
(C) $46 \lfloor 7$
(D) $40 \lfloor 7$

11) Number of natural solutions of the equation $xyz = 2^5 \times 3^2 \times 5^2$ is equal to -

(A) 756
(B) 520
(C) 720
(D) 120

12) Number of complex numbers satisfying equation $z^3 = \bar{z}$ & $\arg(z + 1) = \frac{\pi}{4}$ simultaneously is -

(A) 1
(B) 2
(C) 3
(D) 4

13) Area bounded by $-8z^2 - 8\bar{z}^2 + 20z\bar{z} = 144$ is (where z is a complex number) -

(A) π
(B) 6π
(C) 12π
(D) 4π

14) Number of complex numbers z such that $|z| + z - 3\bar{z} = 0$ is equal to

(A) 0
(B) 1
(C) 2
(D) 3

15) C_0, C_1, C_2 ... C_n are the binomial coefficients in the expansion of $(1 + x)^n$.

Statement-I : $(C^0 + C_1)(C_1 + C_2) \ldots (C_{n-1} + C_n) = k\, C_1C_2 \ldots C_n$, then k is equal to $\frac{(n+1)^n}{(n+1)!}$.
Statement-II : ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.

(A) Statement-I is true but Statement-II is false
(B) Both Statement-I and Statement-II are false

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(C) Both Statement-I and Statement-II are true

(D) Statement-I is false but Statement-II is true

16) Given a set of 100 cards numbered as {00, 01, 02 ... 99}. One card is randomly selected from this set. Let A denotes event that sum of digits on card is 9 and B denotes event that product of digits on card is 0, then $P\left(\frac{A}{B}\right)$ is -

(A) $\frac{1}{9}$

(B) $\frac{2}{9}$

(C) $\frac{1}{19}$

(D) $\frac{2}{19}$

17) An insurance agent insured 50 scooters, 40 cars and 60 trucks in a month. Statistics show that respective probabilities of accidents of these vehicles are 0.02, 0.06 and 0.15. One of his insured vehicles applied for a claim. Probability that the vehicle was a scooter is -

(A) $\frac{5}{62}$

(B) $\frac{3}{62}$

(C) $\frac{1}{55}$

(D) $\frac{1}{56}$

18) Let A and B are two independent events of the sample space S such that $P(A) = \frac{3}{4},\ P(A \cap \bar{B}) = \frac{3}{8}$, then $P(\bar{A} \cap \bar{B}) - P(B \cap \bar{A})$ is equal to

(A) 0

(B) $\frac{1}{8}$

(C) $-\frac{1}{2}$

(D) $\frac{1}{4}$

19) Arithmetic Mean of observations $({}^nC_0)^2,\ ({}^nC_1)^2,\ ({}^nC_2)^2 \ldots ({}^nC_n)^2$ will be

(A) $\frac{\lfloor 2n}{\lfloor n \ \lfloor n}$

(B) $\frac{\lfloor 2n}{(\lfloor n)\ 3}$

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(C) $\frac{\lfloor 2n}{\lfloor n \ \lfloor n+1}$

(D) $\frac{\lfloor 2n}{\lfloor n \ (\lfloor n+1)^2}$

20) The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set

(A) is increased by 2

(B) is decreased by 2

(C) is two times the original median

(D) remains the same as that of the original set

SECTION-II

1) z is a complex number satisfying the condition $\left|z-\frac{5}{z}\right|=4$. Sum of greatest & least value of |z| is

2) The area bounded by the rays arg(z – (1 + i)) = 0, arg(z – (2 + i)) = $\frac{\pi}{2}$; $\arg(z-(2+2i))=\pi$ and $\arg(z-(1+2i))=-\frac{\pi}{2}$ is

3) Let A and B are events of an experiment and $P(A)=\frac{1}{4}$, $P(A\cup B)=\frac{1}{2}$, then value of $P(B/A^C)$ is $\frac{1}{k}$, then the value of k is

4) There are 10 engineering colleges and five students A, B, C, D, E. Each of these students got offer from all of these 10 engineering colleges. They randomly choose college independently of each other. The probability that all get admission in different colleges can be expressed as $\frac{a}{b}$ where a & b are co-prime numbers then the value of a + b is

5) If the variance of 1, 2, 2, 3 is k, then 10k is

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	B	B	A	A	B	A	C	C	C	D	C	D	D	B	B	B	C	C	D	A

SECTION-II

Q.	21	22	23	24	25
A.	5	8	1	16	4

PART-2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	B	C	C	A	D	C	C	B	C	C	C	D	A	B	D	C	C	A	A	D

SECTION-II

Q.	46	47	48	49	50
A.	2	2	3	2	2

PART-3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	B	D	D	A	A	A	B	B	B	B	A	A	C	B	D	D	A	A	C	D

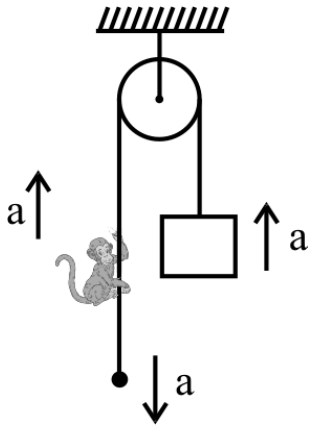
SECTION-II

Q.	71	72	73	74	75
A.	6	1	3	814	5

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SOLUTIONS

PART-1 : PHYSICS



1)

$a_{m/r} = 2a = 2$

So a = 1

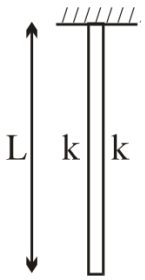
$$10 = \frac{1}{2} \times 1 \times t^2$$

$$t = \sqrt{20} = 2\sqrt{5}$$

2) $\phi = BA\cos\theta = \frac{\mu_0 I}{2\pi} \frac{\pi R^2}{8R^3} \times \frac{\pi r^2}{2} \times \cos 53^\circ$

$M = \frac{\phi}{I} = \frac{3}{160} \frac{\mu_0 \pi r^2}{R}$

3) Moving charge produces both magnetic field and electric field Net magnetic field is due to all current (in side the loop or outside the loop)



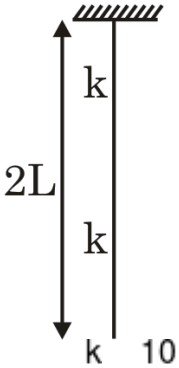
4)

$k_{eq} = 2k = 20$

$\Rightarrow$ k = 10 N/m

$\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{k}$,

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$k'_{eq} = \frac{k}{2} = \frac{10}{2} = 5 \Rightarrow k'_{eq} = 5$

5) $x = \frac{C}{T}$

$\frac{x_1}{x_2} = \frac{T_2}{T_1} = \frac{(273-173)}{(273-73)}$

$\frac{0.0060}{x_2} = \frac{100}{200}$

$x_2 = 0.0120$

6) $A = \frac{F_0/m}{\sqrt{\omega^2 - r^2}}$

7) Let v be the speed of particle at B, just when it is about to loose contact.
From application of Newton's second law to the particle normal to the spherical surface.

$\frac{mv^2}{r} = mg \sin \beta$ ...(1)

Applying conservation of energy as the block moves from A to B..

$\frac{1}{2}mv^2 = mg (r \cos \alpha - r \sin \beta)$ ...(2)

Solving 1 and 2 we get

$3 \sin \beta = 2 \cos \alpha$

8) $W_{spring} = \frac{1}{2}k(a^2 - b^2)$

$W_{spring} + W_{fr} = 0$

$\frac{1}{2}k(a^2 - b^2) - \mu mg(a + b) = 0$

$\mu = \frac{k(a-b)}{2mg}$

9) P → $\tau = 2 \times 10^{-6} \times 2 \times 10^6 = 4$ sec.

Q → $\tau = 2 \times 3 = 6$ sec

R → $\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{3L}{2} \times \frac{2L}{3}}} = 100$

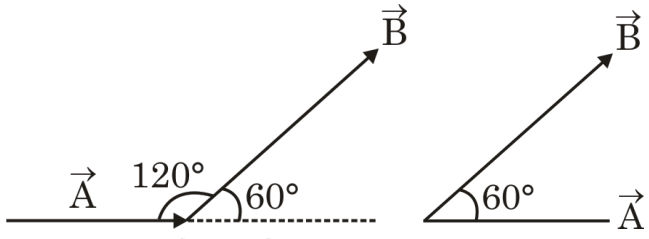
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$S \rightarrow \frac{I(0)}{I(\infty)} = 1.33$

10)

$i = \frac{\varepsilon}{R_{eq}} = \frac{\varepsilon}{2R}$

11)



Angle between $\vec{A}$ and $\vec{B}$, $\theta = 60°$

Angle between $\vec{A}$ and $\vec{A} - \vec{B}$

$$\tan\alpha = \frac{B\sin\theta}{A - B\cos\theta} = \frac{B\sqrt{\frac{3}{2}}}{A - B \times \frac{1}{2}2}$$

$$\tan\alpha = \frac{\sqrt{3}B}{2A - B}$$

12)

Given $\lambda = km^p v^q h^k$. The dimensions of right hand side and left hand side terms should be equal.

So, $\left[M^0LT^0\right] = [M]^p\left[LT^{-1}\right]^q\left[ML^2T^{-1}\right]^r$

or $\left[M^0LT^0\right] = \left[M^{p+r}\right]\left[L^{q+2r}\right]\left[T^{-q-r}\right]$

Now, comparing powers M, L and T, we get

p + r = 0 (i), q + 2r = 1 (ii), – q – r = 0 (iii)

After solving, p = –1, q = –1 and r = 1.

Putting these values, we get

$\lambda = k\frac{h}{mv}$, which is the required relation.

13) $V_C = V_0 = i_0 X_C$

$$V^0 = \frac{i_0}{\omega C} = \frac{V}{\omega C Z}$$

$$V^0 = \frac{V}{\omega C\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

ω is large

$$\frac{V_0}{V} \propto \omega^{-2}$$

14) ϕ = constant

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$\mu_0 n^2 Al = L$

$$\frac{\mu_0 N^2 A}{\ell} = L$$

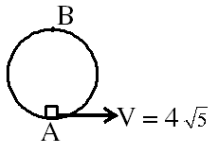
Li = constant

$$\frac{\mu_0 N^2 A}{L_0} i_0 = \frac{\mu_0 N^2 A i_1}{L_1}$$

$$i_0 = \frac{L_1 i_0}{L_0}$$

15)

(P)



Applying Energy conservation between A & B

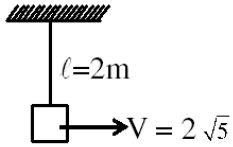
$\frac{1}{2} m(4\sqrt{5})^2 = mg\,(4) + \frac{1}{2}mv^2$

$\Rightarrow 40 = 40 + \frac{1}{2}V^2 \Rightarrow \boxed{V = 0}$

so, Block will just leave the circular path at Highest point & stands to fall.

so $\boxed{N = 0}$ ..... (2)

(Q)

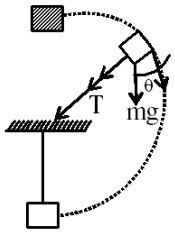


for complete circular motion $v > \sqrt{4gl} \Rightarrow v > \sqrt{80}$

But there $v = 2\sqrt{5} = \sqrt{20}$

so do not complete circular motion

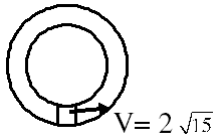
Now, at Highest point, when Block



Reaches, its velocity becomes zero and so it starts to fall. Then mg sin θ

$\Rightarrow \frac{mv^2}{r}$ $\boxed{T = 0}$

(R)

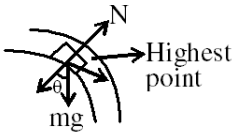


for complete circle,

$V > \sqrt{4gR} \Rightarrow V > \sqrt{80}$, given $v = \sqrt{60}$

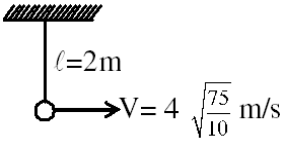
Block will not complete

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Clearly, N = mg cos θ

so, Normal is Radially outward. (4)



(S)

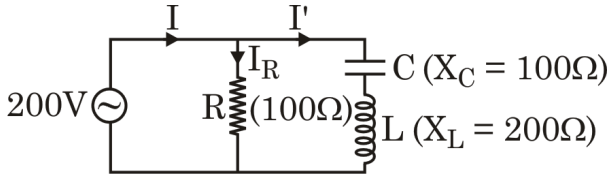
for complete circular motion,

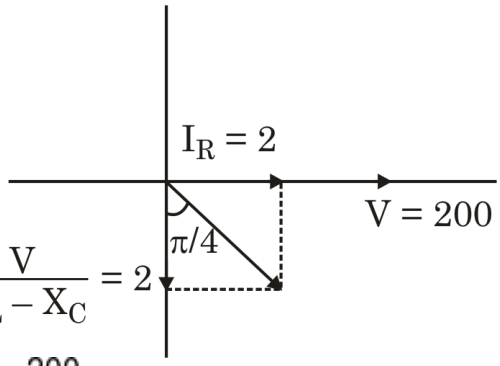
$v > \sqrt{4g\ell} \Rightarrow v > \sqrt{80}$

Here $v = 4\sqrt{\frac{75}{10}} = \sqrt{16 \times \frac{15}{2}} = \sqrt{120}$

so, Here Rod will complete circular motion. (1)

16)





$$I' = \frac{V}{X_L - X_C} = 2$$

$$I_R = \frac{V}{R} = \frac{200}{100} = 2A$$

$$I' = \frac{V}{X_L - X_C} = \frac{200}{100} = 2A$$

$$I = \sqrt{I_R^2 + I'^2} = 2\sqrt{2}\ \text{Amp}$$

17) In LC oscillation

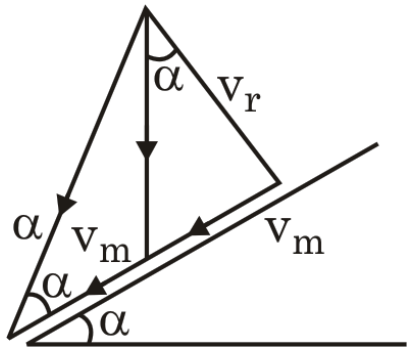
$$f = \frac{1}{2\pi\sqrt{LC}}$$

$q = C\varepsilon \cos\omega t$

$i = C\varepsilon \sin\omega t$

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18)



$\tan\alpha = v_m/v_r$

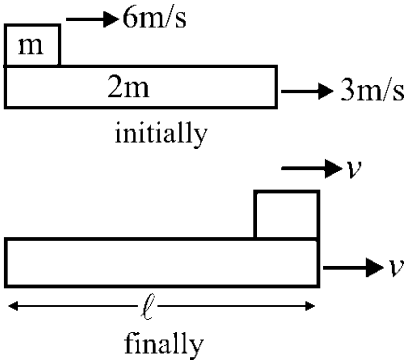
$\tan\alpha = v_r / 2v_m$

$\tan^2\alpha = 1/2$

$\tan\alpha = 1/\sqrt{2}$

19) $t = \dfrac{2u\sin\alpha}{g\cos 37^\circ} = \dfrac{u\cos\alpha}{g\sin 37^\circ}$

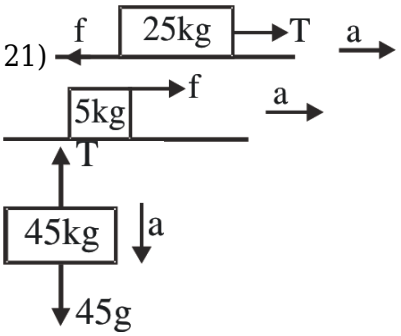
20) $W_{friction} = \Delta KE$



$-\mu mg\ell = \dfrac{1}{2}m3v^2 - \dfrac{1}{2}m6^2 - \dfrac{1}{2}2m3^2$

$\Rightarrow v = 4$ m/s

21)



45g - T = 45 a

T - f = 25 a

f = 5a

45 g = 75 a

$a = \dfrac{45\,g}{75} = 6$

$\Rightarrow f = 5(6) = 30$ N < 75 N

$f_{max} = \mu(25g) = 75$ N

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So, f is static = 30 N

22)

$5\pi = \frac{1}{2} \times 10 \times 1\alpha$

$\Rightarrow \alpha = \pi$ rad/s^2

$\theta = \frac{1}{2} \times \alpha t^2 = 4.5\ \pi$ rad.

$\omega = \alpha t = 3\pi$ rad/s

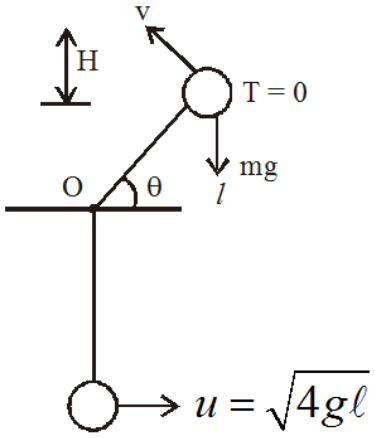
$\alpha' = -\pi \Rightarrow$ comes to rest in $0 = 3\pi - \pi t \Rightarrow 3$sec

$\theta' = 3\pi \times 3 - \frac{1}{2} \times \pi \times 32 = 4.5\pi$

In rest 4 sec $\theta'' = \frac{1}{2} \times \alpha \times t^2 = 8\pi$ rad

23) $mg\cos(90-\theta) = \frac{mv^2}{\ell}$ .....(1)

$\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mg\ell(1+\cos\theta)$ .....(2)



solving, $\sin\theta = \frac{2}{3}$

for projectile motion, $H = \frac{v^2\sin^2(90-\theta)}{2g}$

$H = \frac{5\ell}{27}$

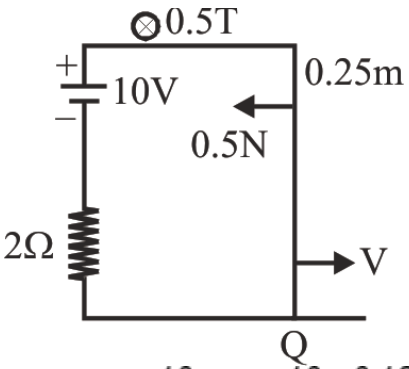
H_{max} for ground $= \ell + \ell\sin\theta + \frac{5\ell}{27} = \frac{50\ell}{27}$

for □ = 54cm, H_{max} = 1 m

24)

Induced e.m.f. = B□v = 0.125 V

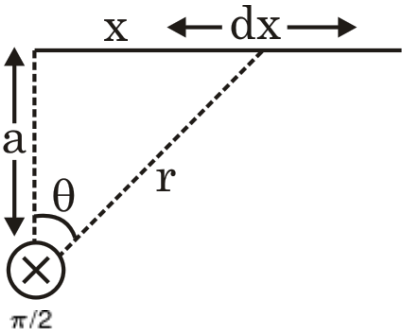
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current $l = \frac{10-e}{R} = \frac{10-0.125V}{2}$

Force $Bl\square = 0.5\left(\frac{10-0.125V}{2}\right)0.25 = 0.5N(given)$

Solving V = 16 m/s.



25) $dF = I_2 \times dx \times \frac{u_0 I}{2\pi r}$

$F = \int_0^{\pi/2} dF$, $\tan\theta = \frac{x}{a}$, $dx = a\sec^2\theta\, d\theta$,

$r = a\sec\theta$

PART-2 : CHEMISTRY

27) Acetaldehyde is better acceptor than acetophenon

30) $\frac{(t_{1/2})_I}{(t_{1/2})_{II}} = \left[\frac{10}{20}\right]^{n-1}$

$\frac{20}{40} = \left(\frac{1}{2}\right)^{n-1}$

$1 = n - 1$

$n = 2$

31)

$P_T = P° + x = P° + kt$

33) Factual

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35) Theory based.

38) Bond order of O_2 and NO^- have two (2) unpaired electrons & both are 16-electron system

hence B.O. = $\frac{10-6}{2} = 2$

41)

(A) $[Fe(SCN)_6]^{3-}$ $[Fe(C_2O_4)_3]^{3-}$

Fe^{3+} Fe^{3+}

Since both are weak field ligands but $C_2O_4^{2-}$ is an stronger field ligand compare to SCN^-. There for splitting will be more in case of $[Fe(C_2O_4)_3]^{3-}$ Correct order

$[Fe(SCN)_6]^{3-} < [Fe(C_2O_4)_3]^{3-}$

(B) In case of Co^{3+} all except $[Co(F)_6]^{3-}$ and $[Co(H_2O)_3F_3]$ will form inner orbital complex of d^2sp^3 since Δ_0 is increased.

Δ_0 will be more for $[Co(H_2O)_6]^{3+} > [Co(H_2O)_3F_3]$

(C) $[NiF_6]^{2-} > [NiF_6]^{4-}$

Δ_0 is given in correct order since Ni^{4+} will have more splitting compare to Ni^{+2}.

(D) $Fe^{3+} > Fe^{2+}$ since as charge increases there is increase in splitting energy Δ_0.

So option (C) is correct.

43) $C_2O_4^{-2}$ replaces ligand on cis position.

46) No. of times increase in rate = $(T.C)^{\frac{\Delta T}{10}}$

$\Rightarrow 512 = (T.C.)^{\frac{100-10}{10}}$

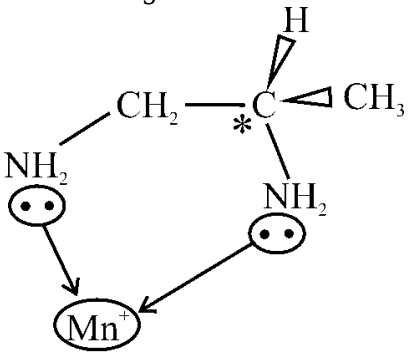
$\Rightarrow 512 = (T.C.)^9$

$\Rightarrow T.C. = 2$

47) $t_{1/2} \propto (a)^{1-n}$

48) 1, 2, 6 is having same number of carbon atoms in reactant and final products.

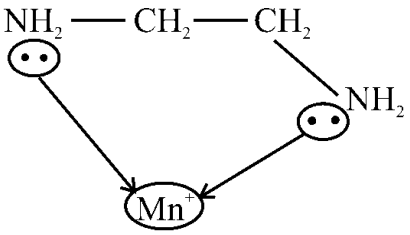
50) Total ligands have five membered ring and also contains chiral centre i.e. "pn" & "bn"02



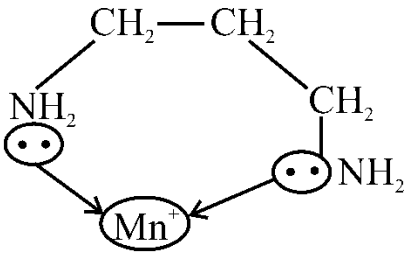
"pn" = Propylene diamine :

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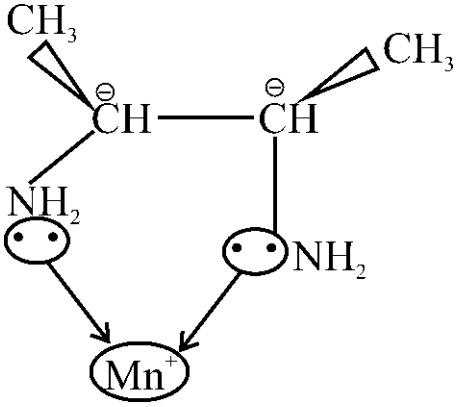
en = Ethylenediamine



"tn" :- trimethylenediamine



bn : - butylenediamine :



PART-3 : MATHEMATICS

51) Let $Q(x) = x^2 - 2\lambda x + 6\lambda - 41$
As $Q(x) < 0 \ \forall \ x \in (1, 6]$, so
$Q(1) \le 0$ ....(1)
& $Q(6) < 0$ ....(2)
must be satisfied simultaneously
So, (1) $\cup$ (2) $\Rightarrow \lambda \in \left(\frac{-5}{6}, 10\right]$

52) $y^2 - 5y < \frac{2}{(x+1)^2 + 3}$
$y^2 - 5y \le 0$
$\Rightarrow 0 \le y \le 5$ y = 0, 1, 2, 3, 4, 5

53) A.M. $\ge$ G.M.
$\Rightarrow \left(\frac{\tan A + 1 + \frac{1}{\tan A}}{3}\right) \geqslant \left(\tan A.1.\frac{1}{\tan A}\right)^{1/3}$

54) Given $a, d \in I^+$ and $a + 2d = 8 \Rightarrow a = 8 - 2d$
also, $191 < \frac{15}{2}(2a + 14d) < 196$

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$\Rightarrow \frac{191}{15} < a + 7d < \frac{196}{15}$

$\Rightarrow \frac{191}{15} < 8 + 5d < \frac{196}{15} \Rightarrow \frac{71}{75} < d < \frac{76}{75}$ $\square$ $d \in I^+$

$\square$ d = 1 and a = 6

$\square$ sum of first 20 terms $= \frac{20}{2}[2a + 19d]$

$= 10[12 + 19] = 310$

55) $V_2 = 3V_1 - 2V_0 = 5$

$V_3 = 3V_2 - 2V_1 = 9$

$V_4 = 3V_3 - 2V_2 = 17$

$V_5 = 3V_4 - 2V_3 = 33$

$V_6 = 3V_5 - 2V_4 = 65$

56) $T_{(r+1)} = {}^{400}C_r 3^{(50-\frac{r}{8})} \quad 5^{\frac{r}{3}}$

r = 0, 24, ..., 24 × 16

Total terms = 17

57) $\int_0^1 \left(1 + x + x^2\right)^{20} (2x + 1)\,dx$

$= \frac{a_0}{1} + \frac{a_1}{2} + \ldots + \frac{a_{41}}{42}$

58) Total number of words in which RR together $= \frac{\lfloor 6}{\lfloor 2 \lfloor 2} = 180$

Let A = number of words in which RR and together and

B = number of words in which RR and EE together.

$\square$ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= \frac{\lfloor 5}{\lfloor 2} + \frac{\lfloor 5}{\lfloor 2} - \lfloor 4 = 96$

$\square$ By inclusion and exclusion principle required number of words = 180 – 96 = 84.

59) $P(x) = 2x^2 - 10px + 7p - 1$

(A) 

$\Rightarrow P(1) > 0$ & $P(-1) > 0$

& $-1 < \frac{-b}{2a} < 1$ & $D \geq 0$

$-3p + 1 > 0$ & $17p + 1 > 0$

& $-1 < \frac{10p}{2 \times 2} < 1$ & $p \in R$

$p < \frac{1}{3} \quad p > -\frac{1}{17} \quad -\frac{2}{5} < p < \frac{2}{5}$

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$\boxed{p \in \left(-\frac{1}{17}, \frac{1}{3}\right)}$

(B) $P(1).P(-1) \le 0$

$(-3p + 1)(17p + 1) < 0$

$\left(p - \frac{1}{3}\right)\left(p + \frac{1}{17}\right) \geqslant 0 \Rightarrow \boxed{p \leqslant \frac{-1}{17} \text{ or } p \geqslant \frac{1}{3}}$

(C)



$f(1) > 0$ & $1 < -\frac{b}{2a}$

$p < \frac{1}{3}$ & $1 < \frac{5p}{2}$

$\frac{2}{5} < p \Rightarrow p \in \phi$

(D) $P(1) < 0$

$\boxed{p > \frac{1}{3}}$

60) Required ways = n(B_1, B_2 separated)
-n(B_1, B_2 separated and G_1, G_2 together)
$= {}^8P_2 \times \lfloor 7 - {}^7P_2 \times \lfloor 6 \times \lfloor 2 = 44 \lfloor 7$

61) $xyz = 2^5 \times 3^2 \times 5^2$

let $x = 2^{a_1}3^{b_1}5^{c_1}$

$y = 2^{a_2}3^{b_2}5^{c_2}$

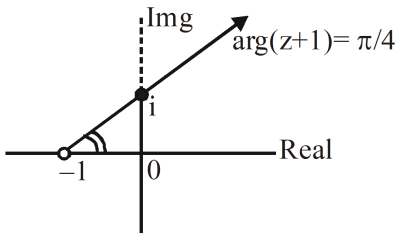
$z = 2^{a_3}3^{b_3}5^{c_3}$

then $xyz = 2^{a_1+a_2+a_3}3^{b_1+b_2+b_3}5^{c_1+c_2+c_3}$

$\Rightarrow a_1 + a_2 + a_3 = 5$

$b_1 + b_2 + b_3 = 2$

$c_1 + c_2 + c_3 = 2$



62)

$z^3 = \bar{z} \Rightarrow |z| = 0$ or 1

$|z| = 0 \Rightarrow z = 0$

$|z| = 1 \Rightarrow z_4 = 1 \Rightarrow z = \pm 1, \pm i$

only z = i satisfies $\arg(z + 1) = \frac{\pi}{4}$

63) Let z = x + iy

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$-8((x+iy)^2+(x-iy)^2)+20(x^2+y^2)=144$
$-8(2(x^2-y^2))+20(x^2+y^2)=144$
$4x^2+36y^2=144$
$\frac{x^2}{36}+\frac{y^2}{4}=1$
$a=6, b=2$
Area $=\pi ab=12\pi$

64) Let $z=x+iy$
$\sqrt{x^2+y^2+x+iy-3x+3iy}=0$
$y=0$ and $\sqrt{x^2+y^2}=2x$
$x=0$
(0, 0) is only possibility

65) ${}^nC_r+{}^nC_{r-1}={}^{n+1}C_r=\frac{(n+1)!}{r!(n+1-r)!}=\frac{(n+1)n!}{r!(n-r)!(n+1-r)}={}^nC_r\frac{n+1}{n+1-r}$
So $\prod_{r=1}^{n}({}^nC_{r-1}+{}^nC_r)=\prod_{r=1}^{n}{}^nC_r\frac{n+1}{n+1-r}=\left(\frac{n+1}{n}\times\frac{n+1}{n-1}\times\frac{n+1}{n-2}\times\ldots\frac{n+1}{2}\times\frac{n+1}{1}\right)C_1C_2$
$\ldots C_n$
$=\frac{(n+1)^n}{n!}C_1C_2\ldots C_n$
so $k=\frac{(n+1)^n}{n!}$
So Statement-I is false and Statement-II is true.

66) $P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$
$A\cap B=\{90,09\}$
$B=\{00, 01 \ldots 10, 20, 30 \ldots, 90\}$
$\therefore P\left(\frac{A}{B}\right)=\frac{2}{19}$

67) By Baye's Theorem
$P\left(\frac{\text{scooter}}{\text{claims}}\right)=\frac{\frac{50}{150}\times.02}{\frac{50}{150}\times.02+\frac{40}{150}\times.06+\frac{60}{150}\times.15}=\frac{5}{62}$

68) $\because P(A)=\frac{3}{4}$ and $P(A\cap\bar{B})=\frac{3}{8}$
$\Rightarrow P(A).P(\bar{B})=\frac{3}{8}$
$\Rightarrow P(B)=P(\bar{B})=\frac{1}{2}$
Now $P(\bar{A}\cap\bar{B})-P(B\cap\bar{A})$
$=P(\bar{A}).(\bar{B})-P(B).(\bar{A})$
($\because$ A and B are independent)

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$$= \frac{1}{4}\cdot\frac{1}{2} - \frac{1}{2}\cdot\frac{1}{4} = 0$$

69) $$\text{A.M.} = \frac{({}^nC_0)^2 + ({}^nC_1)^2 + \ldots + ({}^nC_n)^2}{(n+1)}$$

$$= \frac{{}^{2n}C_n}{n+1} = \frac{\lfloor 2n}{\lfloor n \ \lfloor n \ (n+1)} = \frac{\lfloor 2n}{\lfloor n \ \lfloor n+1}$$

70) Obvious

71) $$\left||z| - \left|\frac{5}{z}\right|\right| \leq \left|z - \frac{5}{z}\right| \Rightarrow \left||z| - \frac{5}{|z|}\right| \leq 4$$

Let $|z| = t$

$-4 \leq t - \frac{5}{t} \leq 4 \Rightarrow -4t \leq t^2 - 5 \leq 4t$

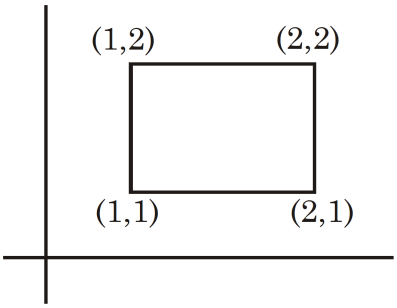
Now, $t^2 + 4t - 5 \geq 0$ $\quad t^2 - 4t - 5 \leq 0$

$(t+5)(t-1) \geq 0$ $\quad (t-5)(t+1) \leq 0$

$\Rightarrow t \in [1, \infty)$ $\quad t \in [0, 5]$ (as t is positive)

$\therefore t \in [1, 5]$

72)



73) $$P(A) = \frac{1}{4}$$

$$P(A \cup B) = \frac{1}{2}$$

$$P(B/A^C) = \frac{P(A^C \cap B)}{P(A^C)} = \frac{P(B) - P(A \cap B)}{P(A^C)}$$

$$= \frac{P(A \cup B) - P(A)}{1 - \frac{1}{4}} = \frac{\frac{1}{2} - \frac{1}{4}}{1 - \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

74) Total no. of cases = 10^5

All students want different colleges so no. of cases = ${}^{10}C_5 \times 5!$

Probability = $$\frac{{}^{10}C_5 \times 5!}{10^5} = \frac{189}{625} = \frac{a}{b}$$

$a + b = 189 + 625 = 814$

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75) x_i x_i^2

1 1

2 4

3 9

$\Sigma x_i = 8$; $\Sigma x_i^2 = 18$

Variance = $\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$

$= \frac{18}{4} - \left(\frac{18}{4}\right)^2 = \frac{1}{2}$

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